HUX / V1-PROOF / ELECTRICAL

# Complete system wiring by location

Rear elevation of the R05 body. Robot LEFT (near wall) is at the left. Each board sits where it mounts. Ports are connector positions. Every harness conductor is drawn.

**SOURCE-CHECKED PLAN**
HARNESS / BENCH RELEASE OPEN

**LEFT LEG · NEAR WALL SIDE**
fixture buttons, pack lead, left gearmotor

**READING THE SHEET**
Wire IDs → searchable conductor register.
Bundle cut lengths → EL-06 and the harness register.
Port labels: connector position / pin.
Dots connect. Gaps at crossings do not.
Dashed boards need part selection and bench checks.
**BLACK wheel lead is switched OUT2, not ground.**
XH = JST XH 2.50 mm. XT30 = 12 V power path.
1x6 0.1 in = the Pololu pigtail housing.

**CTRL DECK · Z +170 · near half: IMU · far half: Pico 2**

**U1 / SparkFun LSM6DSO**
3.3 V · main SPI pads · open the ADDR jumper for SPI · EL-02

JMAIN.5 / SDO — W002 — PH-C.1 / GP12
JMAIN.6 / CS — W003 — PH-C.2 / GP13
JMAIN.1 / GND — W004 — PH-C.3 / GND
JMAIN.4 / SCL — W005 — PH-C.4 / GP14
JMAIN.3 / SDA / SDI — W006 — PH-C.5 / GP15
JINT.2 / INT1 — PH-C.6 / GP16
JINT.1 / INT2
JMAIN.2 / 3V3 — W008 — SP1 splice — PH-F.1 / 3V3(OUT)

JMAIN: B6B-XH-A on the six main pads.
JINT: B2B-XH-A on INT2 / INT1.
INT2 stays empty.
Ports are in cable order; the position number gives the pad.

**P1 / Pico 2**
RP2350 · 3.3 V GPIO · proposed XH groups, physical layout open · GP25 LED onboard

PH-F.2 / 3V3_EN
PH-F.3 / GND
PH-F.4 / VSYS
PH-G.1 / GP0
PH-G.2 / GP1
PH-G.3 / GND

Headers: PH-A p4–p7, PH-A2 p8–p10, PH-B p11–p15, PH-C p16–p21, PH-D p24–p29, PH-E p31–p34, PH-F p36–p39, PH-G p1–p3.
PH-C crosses both Pico rows. Resolve the interface before header removal.
NC: ADC_VREF p35, RUN p30, 3V3_EN p37, GP17 p22, GP19–GP21.
EL-02 shows the physical board and the IMU pins.

P1 / Pico 2

USB.1 / VBUS — W041 — CABLE.1 / VBUS
USB.2 / D+ — W042 — CABLE.2 / D+
USB.3 / D- — W043 — CABLE.3 / D-
USB.4 / GND — W044 — CABLE.4 / GND

**RIGHT HARNESS TRUNK**
down the front face of the body

PH-A.1 / GP2
PH-A.2 / GP3
PH-A.3 / GP4
PH-A.4 / GP5
PH-A2.1 / GND
PH-A2.2 / GP6 — W049
PH-A2.3 / GP7 — W050
PH-B.1 / GP8 — W054
PH-B.2 / GP9 — W055
PH-B.3 / GND — W053
PH-B.4 / GP10
PH-B.5 / GP11 — W051
PH-D.1 / GP18 — W056
PH-D.2 / GP19
PH-D.3 / GP20
PH-D.4 / GP21
PH-D.5 / GND — W058
PH-D.6 / GP22
PH-E.1 / GP26 — W052 — W057
PH-E.2 / GP27
PH-E.3 / AGND
PH-E.4 / GP28

**RIGHT LEG · FAR WALL**
USB cable, right gearmotor

**JUSB / Laptop USB**
CABLE.1 / VBUS
CABLE.2 / D+
CABLE.3 / D-
CABLE.4 / GND

Micro-USB cable to the front of the Pico 2.

**BUS DECK · Z +120 · near half: power distribution · far half: spare cassette position**

JPICO.1 / VSYS — JPICO.2 / GND

**PDB / Power distribution**
From the unswitched fused bus. D5 Schottky → VSYS.
Logic stays on through a kill. 5 V to JLEG.
5 V regulator · part TBD

J5V.1 / 5V
J5V.2 / GND
JIN.1 / BUS
JIN.2 / GND

**SPARE CASSETTE POSITION**
Was the servo bus and the encoder conversion. No board in V1-PROOF.
The encoder dividers sit on JLEG.

W011
W010

W046
W047
W048
W045
W009

**PWR DECK · Z +60 · near half: kill chain · far half: driver board**

JBUS.1 / BUS
JBUS.2 / GND
JCTL.1 / nSLEEP
JCTL.2 / spare
J3V3.1 / 3V3
J3V3.2 / spare

**KC / Kill chain**

SWK / KILL
LEADS.1 / NO — W037 — JKILL.1 / 3V3K
LEADS.2 / COM — W038 — JKILL.2 / OFF node
red mushroom · momentary NO · fixture

SWA / ARM
LEADS.1 / NO — W039 — JARM.1 / 3V3K
LEADS.2 / COM — W040 — JARM.2 / ON node
green · momentary NO · fixture

BAT / 3S pack
LEADS.1 / + — FUSE.1 — W035 — XPACK.1 / +
LEADS.2 / - — FUSE.2 — XPACK.2 / -
on the tray · XT30 · TBD
inline holder · value TBD

The pack sits on the tray at the bottom of the body. It is drawn here so that its lead to the kill chain is clear.
Both buttons are on the bench fixture, not on the robot.

Pololu 2813 + 74HC14 · EL-04.
XPACK: fused pack in.
XRAIL: 2813 VOUT = the switched rail.
LD1117V33 gate logic from the bus.
100k / 22k rail divider → RAIL_ADC.
KILL → OFF. ARM → ON via 4.7k, clamped by HB_OK.
Not ordered or tested. K0–K6 open.

JHB.1 / HB
JHB.2 / RAIL_ADC
JHB.3 / AGND
XRAIL.1 / + — W014 — XRAIL.1 / +
XRAIL.2 / - — W015 — XRAIL.2 / -

**DB / Driver board**
DL, DR: Pololu 4035 / DRV8874 carriers on one 52 x 38 mm board · EL-03, EL-05
10 kΩ nFAULT pull-ups to 3V3. 1 kΩ CS series.
VREF divider ≈ 2.8 V (≈ 2.5 A). PMODE, IMODE to GND. VM NC.
nSLEEP from GP10 to both carriers.
EN low = brake. nSLEEP low = outputs off.
OUT1 / OUT2 go to JLEG, not to the motor directly.

JSIG-L.1 / EN
JSIG-L.2 / PH
JSIG-L.3 / nFAULT
JSIG-L.4 / CS out
JSIG-L.5 / GND
JSIG-R.1 / EN
JSIG-R.2 / PH
JSIG-R.3 / nFAULT
JSIG-R.4 / CS out
JSIG-R.5 / GND

JML.1 / OUT1
JML.2 / OUT2
JMR.1 / OUT1
JMR.2 / OUT2
W060
W061
W062
W063

W016

**UNDER THE DECKS · leg junction board under the PWR deck · 3S pack on the tray at Z −35**

J5V.1 / 5V
J5V.2 / GND
JPL.1 / OUT1
JPL.2 / OUT2
JPR.1 / OUT1
JPR.2 / OUT2
JENC.1
JENC.2
JENC.3
JENC.4
JENC.5

**JLEG / Leg junction**
Accepted 2026-10-17 · under the PWR deck · 1x6 pigtail headers · 2.2 kΩ / 3.3 kΩ dividers
Motor leads pass to JPL / JPR. Encoder A / B 5 V → 3.0 V → JENC.
Board layout, mount detail and the service loop for later leg motion remain open.

**ML / Pololu 4752 · LEFT WHEEL**
ML
12 V · 30:1 · encoder

PIGTAIL.1 / RED — W022 — JML_IN.1 / RED
PIGTAIL.2 / BLACK — W023 — JML_IN.2 / BLACK
PIGTAIL.3 / GREEN — W024 — JML_IN.3 / GREEN
PIGTAIL.4 / BLUE — W025 — JML_IN.4 / BLUE
PIGTAIL.5 / YELLOW — W026 — JML_IN.5 / YELLOW
PIGTAIL.6 / WHITE — W027 — JML_IN.6 / WHITE

JMR_IN.1 / RED — W028 — PIGTAIL.1 / RED
JMR_IN.2 / BLACK — W029 — PIGTAIL.2 / BLACK
JMR_IN.3 / GREEN — W030 — PIGTAIL.3 / GREEN
JMR_IN.4 / BLUE — W031 — PIGTAIL.4 / BLUE
JMR_IN.5 / YELLOW — W032 — PIGTAIL.5 / YELLOW
JMR_IN.6 / WHITE — W033 — PIGTAIL.6 / WHITE

**MR / Pololu 4752 · RIGHT WHEEL**
MR
12 V · 30:1 · encoder

HUX / V1-PROOF / ELECTRICAL

# Pico 2 + IMU connections

Component-side views, USB at the top of the Pico. Physical pins and GPIO numbers are distinct.

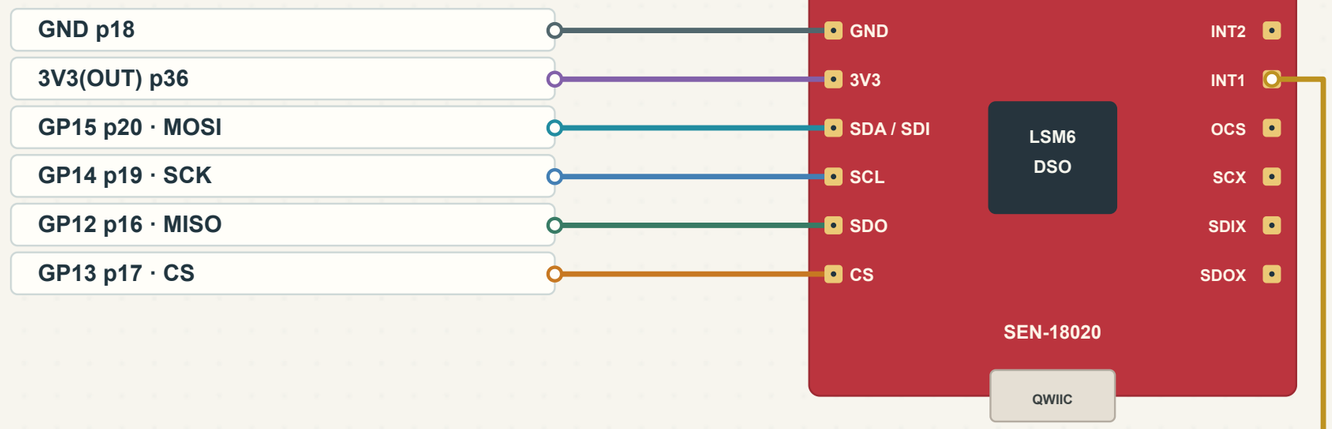
**SOURCE-CHECKED PLAN**
HARNESS / BENCH RELEASE OPEN

## PHYSICAL PIN MAP




p1 is upper left; p40 is upper right.

## SEVEN-WIRE SENSOR HARNESS




**SPI JUMPER SETUP / CHECK RECEIVED BOARD**

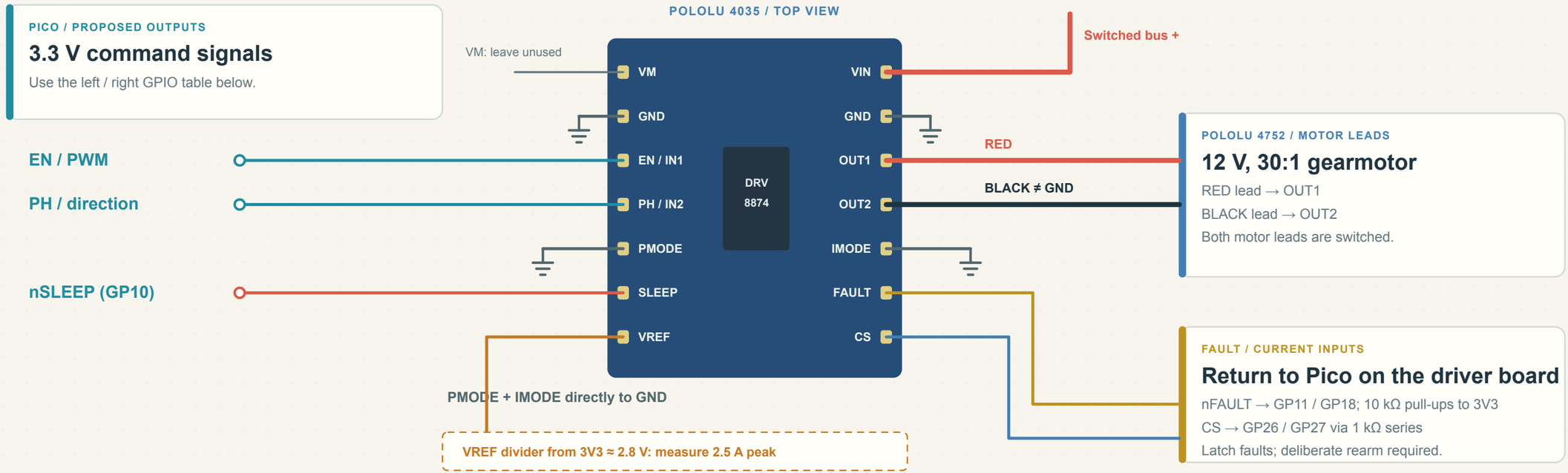
Open the 0x6B / 0x6A address jumper so SDO can drive MISO.
Recommended: open both I2C pull-up traces. Keep SCX / SDIX ground jumpers intact.
Qwiic sockets carry I2C and are unused here. Wire the main SPI pads, not SCX / SDIX.
Power and signal logic are 3.3 V. Inspect board revision and check sensor identity before use.

HUX / V1-PROOF / ELECTRICAL

# Wheel motor + encoder harness

One channel shown; repeat for the other wheel using the assignment table. Colors match the motor leads. The pigtail lands on the JLEG junction board.

**SOURCE-CHECKED PLAN**
HARNESS / BENCH RELEASE OPEN




PICO / LEFT AND RIGHT CHANNELS

## Proposed pin assignments

| Function | Left | Right |
|---|---|---|
| Encoder A | GP2 | GP4 |
| Encoder B | GP3 | GP5 |
| EN / PH | GP6/7 | GP8/9 |
| FAULT | GP11 | GP18 |
| CS / ADC | GP26 | GP27 |

JLEG JUNCTION BOARD / PROPOSED

## 5 V → 3.0 V resistor dividers

Four channels: 2.2 kΩ series, 3.3 kΩ to GND
Bench value from the electronics bench guide.
Measure the high level on GP2 before use.
No direct 5 V signals into Pico GPIO.

GND / GREEN
5 V / BLUE
A / YELLOW
B / WHITE

4752 / ENCODER LEADS

## Six-wire vendor pigtail, 200 mm

GREEN → common GND
BLUE → regulated 5 V
YELLOW → channel A
WHITE → channel B
1x6 0.1 in housing → JLEG header. Verify by color.

**EN low = brake. nSLEEP low = outputs off / coast. GP10 drives both nSLEEP pins.**

2.5 A peak / 1.2 A RMS are Hux qualification targets, not tested settings.

HUX / V1-PROOF / ELECTRICAL

# Power distribution + kill chain

Functional drawing of docs/kill-chain.md: fused pack, latched rail switch, heartbeat gate, logic branch. No part is ordered or tested.

**SOURCE-CHECKED PLAN**
HARNESS / BENCH RELEASE OPEN

BATTERY / TBD
### 3S pack
Up to 12.6 V charged · XT30

F1 / INLINE
### Main fuse
Value and wire gauge open

KC / POLOLU 2813
### Latched rail switch
VIN → VOUT. ON / OFF pulse inputs.

RAIL

DB / ×2
### DRV8874 VIN
Both carriers on the switched rail

BUS → logic

BUS → gate logic

PDB / UNSWITCHED LOGIC BRANCH
### 5 V regulator, part TBD
Fed before the switch: logic stays on through a kill
5 V to the encoders through JLEG

OPEN ITEMS
Fuse value, rail wire gauge and the XT30 fit.
Return-energy clamp: the switch opens the supply path.
ON / OFF priority of the 2813 (stage K2).

KC / HEARTBEAT GATE
### 74HC14 + charge pump
GP22 500 Hz → 220 nF → BAT54S → 1 µF ‖ 100 kΩ
No edges for ≈ 80 ms → /HB_OK high → OFF
Own LD1117V33 3.3 V from the bus

PDB / USB + EXTERNAL POWER
### D5 Schottky → VSYS p39
External 5 V → diode → VSYS
Check the loaded voltage and USB backfeed

VSYS p39

P1 / CONTROLLER
### Pico 2 power + sense
3V3(OUT) p36 → IMU + driver pull-ups
GP28 p34 ← rail divider 100 kΩ / 22 kΩ

KC / BUTTONS ON THE FIXTURE
### KILL and ARM
KILL: 3.3 V → button → diode → OFF
ARM: 3.3 V → button → 4.7 kΩ → ON
ON clamped low while HB_OK is low
Second KILL press does nothing.
Rail returns only on ARM with the heartbeat.

SOFTWARE DISABLE, SEPARATE LAYER
GP10 p14 → both DRV8874 nSLEEP pins directly.
disarm and FAULT set nSLEEP low in the same cycle.
The rail switch is the second, independent layer.
A reset or reboot never turns the rail on.

P1 → KC / SENSE AND HEARTBEAT
### Bundle B7, connector KC.JHB
GP22 p29 → JHB.1 HB
GP28 p34 ← JHB.2 RAIL_ADC
AGND p33 → JHB.3

HB / RAIL_ADC

COMMON GND / RETURNS

All GND symbols join this reference. Route the rail and the pack return away from the sensor and MCU wiring.

Rev C | 17 OCT 2026 | Vendor pinouts + proposed Hux circuits

See docs/wiring-atlas.md and the wire register for sources and open circuit decisions.

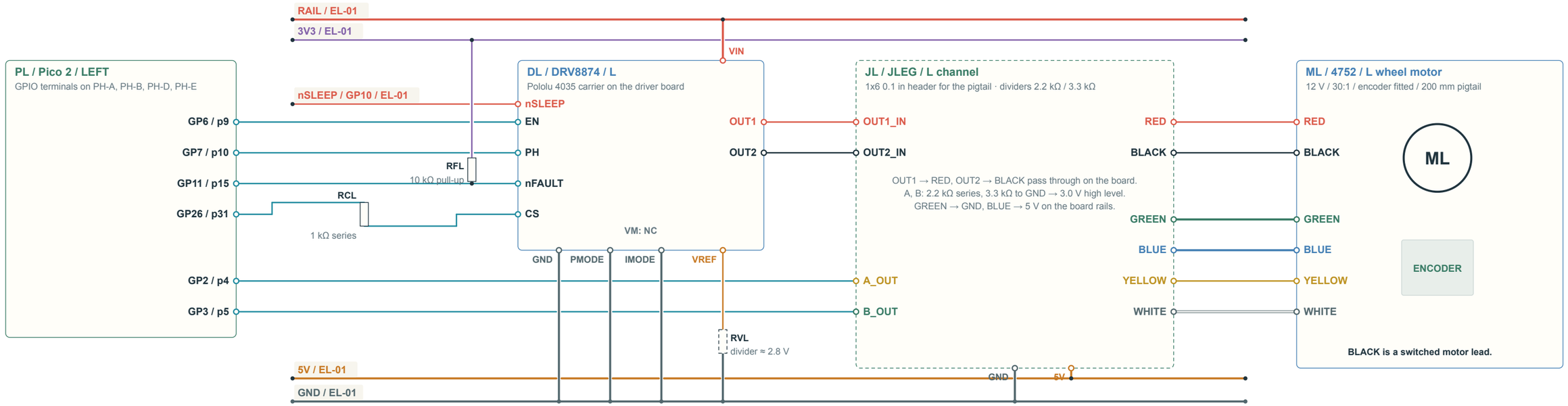
EL-04

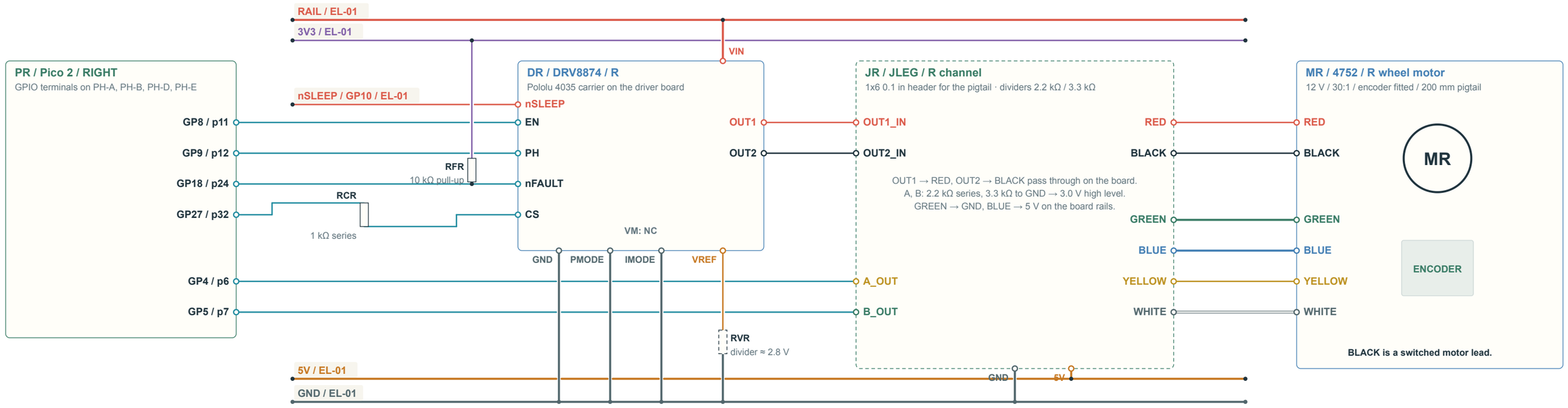
HUX / V1-PROOF / ELECTRICAL

# Both wheel motors — every wire

Two complete channels through the leg junction board. Rails and nSLEEP come from EL-01. Use terminal names here. EL-01 has the conductor register IDs.

**SOURCE-CHECKED PLAN**
HARNESS / BENCH RELEASE OPEN



# Harnesses

Twelve harnesses. Each card shows the two ends, the housings with their position count and the cut length of each branch. Lengths are estimates: cut long, route on the frame, trim, record.

**SOURCE-CHECKED PLAN**
HARNESS / BENCH RELEASE OPEN

## H1 · CTRL deck harness

11 wires · longest cut 335 mm

- P1.PH-C (6) → U1.JMAIN (6) — 95 mm
- P1.PH-F (4) → U1.JINT (2) — 95 mm
- → PDB.JPICO (2) — 330 mm
- → DB.J3V3 (2) — 335 mm

Sensor wires to the IMU, logic power from the PDB, one 3V3 tail down the far side to the driver board. Splice SP1 near the Pico.

## H2 · Right trunk harness

19 wires · longest cut 395 mm

- P1.PH-A (4) → DB.JSIG-L (5) — 335 mm
- P1.PH-A2 (3) → DB.JSIG-R (5) — 335 mm
- P1.PH-B (5) → DB.JCTL (2) — 335 mm
- P1.PH-D (6) → JLEG.JENC (5) — 350 mm
- P1.PH-E (4) → KC.JHB (3) — 395 mm

All Pico I/O down the front face of the far half. Three branches: driver board, leg junction encoders, kill-chain sense.

## H3 · Logic bus

2 wires · longest cut 280 mm

- KC.JBUS (2) — 280 mm → PDB.JIN (2)

Unswitched fused bus up to the 5 V regulator. 22 AWG.

## H4 · Switched rail

2 wires · longest cut 95 mm

- KC.XRAIL (2 · XT30) — 95 mm → DB.XRAIL (2 · XT30)

Pololu 2813 VOUT across the PWR deck to both drivers. XT30, 20 AWG.

## H5 · Encoder supply

2 wires · longest cut 295 mm

- PDB.J5V (2) — 295 mm → JLEG.J5V (2)

5 V and GND down the mid gap to the leg junction board. 22 AWG.

## H6 · Left motor power

2 wires · longest cut 100 mm

- DB.JML (2) — 100 mm → JLEG.JPL (2)

OUT1 / OUT2, straight down. 22 AWG.

## H7 · Right motor power

2 wires · longest cut 100 mm

- DB.JMR (2) — 100 mm → JLEG.JPR (2)

OUT1 / OUT2, straight down. 22 AWG.

## H8 · Left motor pigtail

6 wires · longest cut 200 mm

- ML.PIGTAIL (6 · P6) — 200 mm fixed → JLEG.JML_IN (6 · P6)

Supplied cable, 200 mm, 1x6 0.1 in housing.

## H9 · Right motor pigtail

6 wires · longest cut 200 mm

- MR.PIGTAIL (6 · P6) — 200 mm fixed → JLEG.JMR_IN (6 · P6)

Supplied cable, 200 mm, 1x6 0.1 in housing.

## H10 · Pack lead

3 wires · longest cut 200 mm

- BAT.LEADS (2 · pack leads) — 200 mm → F1.INLINE (2 · inline fuse holder)
- — 200 mm → KC.XPACK (2 · XT30)

Pack to the inline fuse to the kill-chain board. XT30 at the board. 20 AWG.

## H11 · KILL button lead

2 wires · longest cut 600 mm

- SWK.LEADS (2) — 600 mm fixed → KC.JKILL (2)

Fixture lead, 600 mm.

## H12 · ARM button lead

2 wires · longest cut 600 mm

- SWA.LEADS (2) — 600 mm fixed → KC.JARM (2)

Fixture lead, 600 mm.

Housings: green = JST XH 2.50 mm, red = XT30, blue = the Pololu 1x6 0.1 in housing. Grey cells are empty positions.
Cut length = route model x 1.10 + 2 x 15 mm, rounded up to 5 mm. The USB cable and the optional UART0 header are not harnesses. EL-07 has the pin of every position.

# Connector loading

Logical position schedules, not connector face views. XH = JST XH 2.50 mm. Verify position 1 on the actual mating parts. PH-C spans both Pico rows: physical interface unresolved.

**SOURCE-CHECKED PLAN**
HARNESS / BENCH RELEASE OPEN

### P1.PH-A
XH · 4 positions · p4-p7 · Encoder inputs

| | | | |
|---|---|---|---|
| 1 | GP2 | L_ENC_A | H021 → JLEG.JENC.1 (B6) |
| 2 | GP3 | L_ENC_B | H022 → JLEG.JENC.2 (B6) |
| 3 | GP4 | R_ENC_A | H023 → JLEG.JENC.3 (B6) |
| 4 | GP5 | R_ENC_B | H024 → JLEG.JENC.4 (B6) |

### P1.PH-A2
XH · 3 positions · p8-p10 · Encoder ground, left EN / PH

| | | | |
|---|---|---|---|
| 1 | GND | GND | H025 → JLEG.JENC.5 (B6) |
| 2 | GP6 | L_EN | H010 → DB.JSIG-L.1 (B3) |
| 3 | GP7 | L_PH | H011 → DB.JSIG-L.2 (B3) |

### P1.PH-B
XH · 5 positions · p11-p15 · Right EN / PH, nSLEEP, left FAULT

| | | | |
|---|---|---|---|
| 1 | GP8 | R_EN | H015 → DB.JSIG-R.1 (B4) |
| 2 | GP9 | R_PH | H016 → DB.JSIG-R.2 (B4) |
| 3 | GND | GND | H014 → DB.JSIG-L.5 (B3) |
| 4 | GP10 | nSLEEP | H020 → DB.JCTL.1 (B5) |
| 5 | GP11 | L_FAULT | H012 → DB.JSIG-L.3 (B3) |

### P1.PH-C
XH · 6 positions · p16-p21 · SPI1 sensor

| | | | |
|---|---|---|---|
| 1 | GP12 | SPI_SDO | H001 → U1.JMAIN.5 (B1) |
| 2 | GP13 | SPI_CS | H002 → U1.JMAIN.6 (B1) |
| 3 | GND | GND | H003 → U1.JMAIN.1 (B1) |
| 4 | GP14 | SPI_SCL | H004 → U1.JMAIN.4 (B1) |
| 5 | GP15 | SPI_SDI | H005 → U1.JMAIN.3 (B1) |
| 6 | GP16 | IMU_INT1 | H006 → U1.JINT.2 (B1) |

### P1.PH-D
XH · 6 positions · p24-p29 · Right FAULT, heartbeat

| | | | |
|---|---|---|---|
| 1 | GP18 | R_FAULT | H017 → DB.JSIG-R.3 (B4) |
| 2 | GP19 | empty | |
| 3 | GP20 | empty | |
| 4 | GP21 | empty | |
| 5 | GND | GND | H019 → DB.JSIG-R.5 (B4) |
| 6 | GP22 | HB | H026 → KC.JHB.1 (B7) |

### P1.PH-E
XH · 4 positions · p31-p34 · Current sense, rail sense

| | | | |
|---|---|---|---|
| 1 | GP26 | L_CS | H013 → DB.JSIG-L.4 (B3) |
| 2 | GP27 | R_CS | H018 → DB.JSIG-R.4 (B4) |
| 3 | AGND | AGND | H028 → KC.JHB.3 (B7) |
| 4 | GP28 | RAIL_ADC | H027 → KC.JHB.2 (B7) |

### P1.PH-F
XH · 4 positions · p36-p39 · Logic power

| | | | |
|---|---|---|---|
| 1 | 3V3(OUT) | 3V3 | H008 → SP1 (B2) |
| 2 | 3V3_EN | empty | |
| 3 | GND | GND | H030 → PDB.JPICO.2 (B8) |
| 4 | VSYS | VSYS | H029 → PDB.JPICO.1 (B8) |

### P1.PH-G
XH · 3 positions · p1-p3 · Optional UART0 link (no cable in V1-PROOF)

| | | | |
|---|---|---|---|
| 1 | GP0 | empty | |
| 2 | GP1 | empty | |
| 3 | GND | empty | |

### P1.USB
Micro-USB · 4 positions · onboard · Onboard Micro-USB

| | | | |
|---|---|---|---|
| 1 | VBUS | USB_VBUS | H060 → JUSB.CABLE.1 (B19) |
| 2 | D+ | USB_D+ | H061 → JUSB.CABLE.2 (B19) |
| 3 | D- | USB_D- | H062 → JUSB.CABLE.3 (B19) |
| 4 | GND | GND | H063 → JUSB.CABLE.4 (B19) |

### U1.JMAIN
XH · 6 positions · main pads · LSM6DSO main SPI pads

| | | | |
|---|---|---|---|
| 1 | GND | GND | H003 → P1.PH-C.3 (B1) |
| 2 | 3V3 | 3V3 | H007 → SP1 (B1) |
| 3 | SDA / SDI | SPI_SDI | H005 → P1.PH-C.5 (B1) |
| 4 | SCL | SPI_SCL | H004 → P1.PH-C.4 (B1) |
| 5 | SDO | SPI_SDO | H001 → P1.PH-C.1 (B1) |
| 6 | CS | SPI_CS | H002 → P1.PH-C.2 (B1) |

### U1.JINT
XH · 2 positions · INT2, INT1 pads · LSM6DSO interrupt pads

| | | | |
|---|---|---|---|
| 1 | INT2 | empty | |
| 2 | INT1 | IMU_INT1 | H006 → P1.PH-C.6 (B1) |

### DB.JSIG-L
XH · 5 positions · protoboard · Left driver signals

| | | | |
|---|---|---|---|
| 1 | EN | L_EN | H010 → P1.PH-A2.2 (B3) |
| 2 | PH | L_PH | H011 → P1.PH-A2.3 (B3) |
| 3 | nFAULT | L_FAULT | H012 → P1.PH-B.5 (B3) |
| 4 | CS out | L_CS | H013 → P1.PH-E.1 (B3) |
| 5 | GND | GND | H014 → P1.PH-B.3 (B3) |

### DB.JSIG-R
XH · 5 positions · protoboard · Right driver signals

| | | | |
|---|---|---|---|
| 1 | EN | R_EN | H015 → P1.PH-B.1 (B4) |
| 2 | PH | R_PH | H016 → P1.PH-B.2 (B4) |
| 3 | nFAULT | R_FAULT | H017 → P1.PH-D.1 (B4) |
| 4 | CS out | R_CS | H018 → P1.PH-E.2 (B4) |
| 5 | GND | GND | H019 → P1.PH-D.5 (B4) |

### DB.JCTL
XH · 2 positions · protoboard · Shared nSLEEP

| | | | |
|---|---|---|---|
| 1 | nSLEEP | nSLEEP | H020 → P1.PH-B.4 (B5) |
| 2 | spare | empty | |

### DB.J3V3
XH · 2 positions · protoboard · Pull-up supply from the Pico 3V3

| | | | |
|---|---|---|---|
| 1 | 3V3 | 3V3 | H009 → SP1 (B2) |
| 2 | spare | empty | |

### DB.JML
XH · 2 positions · protoboard · Left motor OUT1 / OUT2

| | | | |
|---|---|---|---|
| 1 | OUT1 | L_MOTOR_OUT1 | H037 → JLEG.JPL.1 (B12) |
| 2 | OUT2 | L_MOTOR_OUT2 | H038 → JLEG.JPL.2 (B12) |

### DB.JMR
XH · 2 positions · protoboard · Right motor OUT1 / OUT2

| | | | |
|---|---|---|---|
| 1 | OUT1 | R_MOTOR_OUT1 | H039 → JLEG.JPR.1 (B13) |
| 2 | OUT2 | R_MOTOR_OUT2 | H040 → JLEG.JPR.2 (B13) |

### DB.XRAIL
XT30 · 2 positions · pigtail · Switched rail in

| | | | |
|---|---|---|---|
| 1 | + | RAIL | H033 → KC.XRAIL.1 (B10) |
| 2 | - | GND | H034 → KC.XRAIL.2 (B10) |

### JLEG.JML_IN
P6 · 6 positions · 1x6 male header · Left Pololu pigtail

| | | | |
|---|---|---|---|
| 1 | RED | L_MOTOR_OUT1 | H041 → ML.PIGTAIL.1 (B14) |
| 2 | BLACK | L_MOTOR_OUT2 | H042 → ML.PIGTAIL.2 (B14) |
| 3 | GREEN | GND | H043 → ML.PIGTAIL.3 (B14) |
| 4 | BLUE | 5V | H044 → ML.PIGTAIL.4 (B14) |
| 5 | YELLOW | L_ENC_A5 | H045 → ML.PIGTAIL.5 (B14) |
| 6 | WHITE | | H046 → ML.PIGTAIL.6 (B14) |

### JLEG.JMR_IN
P6 · 6 positions · 1x6 male header · Right Pololu pigtail

| | | | |
|---|---|---|---|
| 1 | RED | R_MOTOR_OUT1 | H047 → MR.PIGTAIL.1 (B15) |
| 2 | BLACK | R_MOTOR_OUT2 | H048 → MR.PIGTAIL.2 (B15) |
| 3 | GREEN | GND | H049 → MR.PIGTAIL.3 (B15) |
| 4 | BLUE | 5V | H050 → MR.PIGTAIL.4 (B15) |
| 5 | YELLOW | R_ENC_A5 | H051 → MR.PIGTAIL.5 (B15) |
| 6 | WHITE | | H052 → MR.PIGTAIL.6 (B15) |

### JLEG.JPL
XH · 2 positions · protoboard · Left motor power to driver

| | | | |
|---|---|---|---|
| 1 | OUT1 | L_MOTOR_OUT1 | H037 → DB.JML.1 (B12) |
| 2 | OUT2 | L_MOTOR_OUT2 | H038 → DB.JML.2 (B12) |

### JLEG.JPR
XH · 2 positions · protoboard · Right motor power to driver

| | | | |
|---|---|---|---|
| 1 | OUT1 | R_MOTOR_OUT1 | H039 → DB.JMR.1 (B13) |
| 2 | OUT2 | R_MOTOR_OUT2 | H040 → DB.JMR.2 (B13) |

### JLEG.JENC
XH · 5 positions · protoboard · Encoder outputs, 3.3 V level

| | | | |
|---|---|---|---|
| 1 | L_A | L_ENC_A | H021 → P1.PH-A.1 (B6) |
| 2 | L_B | L_ENC_B | H022 → P1.PH-A.2 (B6) |
| 3 | R_A | R_ENC_A | H023 → P1.PH-A.3 (B6) |
| 4 | R_B | R_ENC_B | H024 → P1.PH-A.4 (B6) |
| 5 | GND | GND | H025 → P1.PH-A2.1 (B6) |

### JLEG.J5V
XH · 2 positions · protoboard · Encoder supply

| | | | |
|---|---|---|---|
| 1 | 5V | 5V | H035 → PDB.J5V.1 (B11) |
| 2 | GND | GND | H036 → PDB.J5V.2 (B11) |

### KC.XPACK
XT30 · 2 positions · pigtail · Fused pack in

| | | | |
|---|---|---|---|
| 1 | + | PACK_FUSED | H054 → F1.INLINE.2 (B16) |
| 2 | - | GND | H055 → BAT.LEADS.2 (B16) |

### KC.XRAIL
XT30 · 2 positions · pigtail · Switched rail out (Pololu 2813 VOUT)

| | | | |
|---|---|---|---|
| 1 | + | RAIL | H033 → DB.XRAIL.1 (B10) |
| 2 | - | GND | H034 → DB.XRAIL.2 (B10) |

### KC.JBUS
XH · 2 positions · protoboard · Unswitched fused bus to the 5 V regulator

| | | | |
|---|---|---|---|
| 1 | BUS | BUS | H031 → PDB.JIN.1 (B9) |
| 2 | GND | GND | H032 → PDB.JIN.2 (B9) |

### KC.JHB
XH · 3 positions · protoboard · Heartbeat in, rail sense out

| | | | |
|---|---|---|---|
| 1 | HB | HB | H026 → P1.PH-D.6 (B7) |
| 2 | RAIL_ADC | RAIL_ADC | H027 → P1.PH-E.4 (B7) |
| 3 | AGND | AGND | H028 → P1.PH-E.3 (B7) |

### KC.JKILL
XH · 2 positions · protoboard · KILL button

| | | | |
|---|---|---|---|
| 1 | 3V3K | KILL_A | H056 → SWK.LEADS.1 (B17) |
| 2 | OFF node | KILL_B | H057 → SWK.LEADS.2 (B17) |

### KC.JARM
XH · 2 positions · protoboard · ARM button

| | | | |
|---|---|---|---|
| 1 | 3V3K | ARM_A | H058 → SWA.LEADS.1 (B18) |
| 2 | ON node | ARM_B | H059 → SWA.LEADS.2 (B18) |

### PDB.JIN
XH · 2 positions · protoboard · Fused bus in

| | | | |
|---|---|---|---|
| 1 | BUS | BUS | H031 → KC.JBUS.1 (B9) |
| 2 | GND | GND | H032 → KC.JBUS.2 (B9) |

### PDB.JPICO
XH · 2 positions · protoboard · 5 V through D5 to VSYS

| | | | |
|---|---|---|---|
| 1 | VSYS | VSYS | H029 → P1.PH-F.4 (B8) |
| 2 | GND | GND | H030 → P1.PH-F.3 (B8) |

### PDB.J5V
XH · 2 positions · protoboard · 5 V to the encoder junction

| | | | |
|---|---|---|---|
| 1 | 5V | 5V | H035 → JLEG.J5V.1 (B11) |
| 2 | GND | GND | H036 → JLEG.J5V.2 (B11) |

### SWK.LEADS
XH · 2 positions · button · KILL button leads

| | | | |
|---|---|---|---|
| 1 | NO | KILL_A | H056 → KC.JKILL.1 (B17) |
| 2 | COM | KILL_B | H057 → KC.JKILL.2 (B17) |

### SWA.LEADS
XH · 2 positions · button · ARM button leads

| | | | |
|---|---|---|---|
| 1 | NO | ARM_A | H058 → KC.JARM.1 (B18) |
| 2 | COM | ARM_B | H059 → KC.JARM.2 (B18) |

### ML.PIGTAIL
P6 · 6 positions · vendor housing · Pololu 4752 leads, 200 mm

| | | | |
|---|---|---|---|
| 1 | RED | L_MOTOR_OUT1 | H041 → JLEG.JML_IN.1 (B14) |
| 2 | BLACK | L_MOTOR_OUT2 | H042 → JLEG.JML_IN.2 (B14) |
| 3 | GREEN | GND | H043 → JLEG.JML_IN.3 (B14) |
| 4 | BLUE | 5V | H044 → JLEG.JML_IN.4 (B14) |
| 5 | YELLOW | L_ENC_A5 | H045 → JLEG.JML_IN.5 (B14) |
| 6 | WHITE | | H046 → JLEG.JML_IN.6 (B14) |

### MR.PIGTAIL
P6 · 6 positions · vendor housing · Pololu 4752 leads, 200 mm

| | | | |
|---|---|---|---|
| 1 | RED | R_MOTOR_OUT1 | H047 → JLEG.JMR_IN.1 (B15) |
| 2 | BLACK | R_MOTOR_OUT2 | H048 → JLEG.JMR_IN.2 (B15) |
| 3 | GREEN | GND | H049 → JLEG.JMR_IN.3 (B15) |
| 4 | BLUE | 5V | H050 → JLEG.JMR_IN.4 (B15) |
| 5 | YELLOW | R_ENC_A5 | H051 → JLEG.JMR_IN.5 (B15) |
| 6 | WHITE | | H052 → JLEG.JMR_IN.6 (B15) |

Pack leads, the inline fuse and the laptop USB cable are not listed; they are not harness connectors. The three empty Pico positions in PH-D keep GP19–GP21 free for a later revision.